



BAAZ & UDAY JKBOSE

2026 CLASS 11TH

SETS

MATHS

LECTURE-05

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TOPICS TO BE COVERED

1

Intervals as Subsets of $\mathbb{R}$

2

Exercise Discussion 1.3

3

Venn Diagram and Operation on Sets

3

Related problems

$$A \not\subseteq B.$$

Quick Recap:

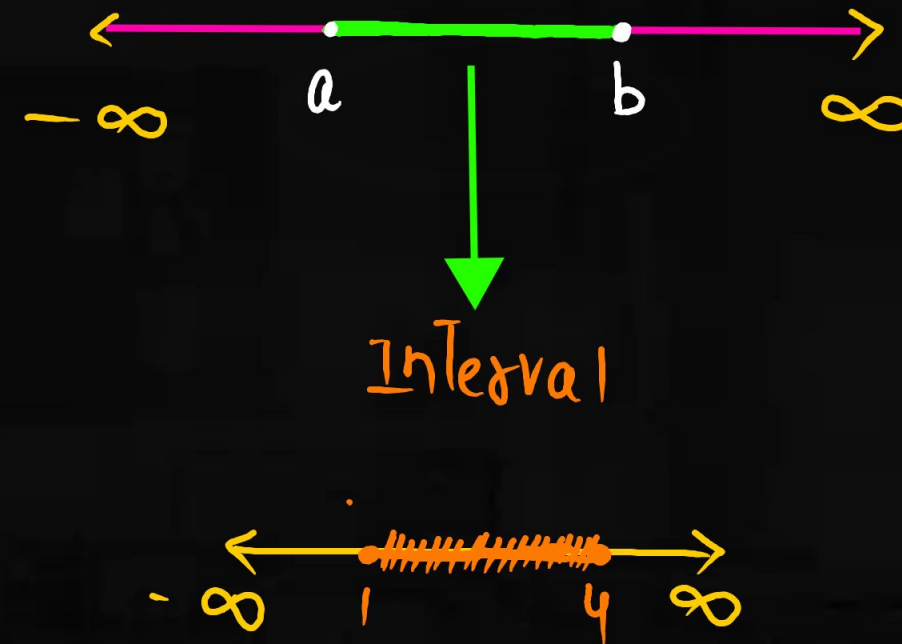
Subset $\rightarrow$ every element of B
is contained in A

$$B = \{1, 2, 4\}, A = \{1, 2, 3, 4, 5\}$$

$$B \subseteq A$$

Intervals

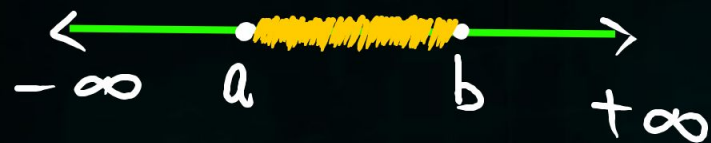
- Continuous part of a Real Line is known as an Interval.
- Here a and b are known as the ends of the Interval.
- Here $b-a$ is known as the length of the Interval.
- Intervals are generally denoted by I



classification of Intervals

on the basis of end points, we have;

(1) open Interval



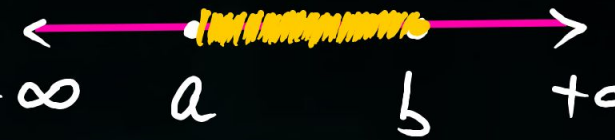
- Both end points are excluded.

$$I = \{x \in \mathbb{R} : a < x < b\} = (a, b)$$

(2) Semi-open / Semi closed:

- only one of the end points are included.

$$I = \{x \in \mathbb{R} : a \leq x < b\} = [a, b)$$


$$I = \{x \in \mathbb{R} : a < x \leq b\} = (a, b]$$


(3) closed Interval:

- Both end points are included.

$$I = \{x \in \mathbb{R} : a \leq x \leq b\} = [a, b]$$


QUESTION

The subset of $\mathbb{R}$ as Intervals $\{x : x \in \mathbb{R}, -12 < x < -10\}$ is

~~1~~ $(-12, -10]$ ✗

~~2~~ $[-12, -10]$ ✗

~~3~~ $(-12, -10)$ ✓

~~4~~ $[-12, -10)$

QUESTION

Which of the following sets is the Universal set of sets $A = \{2, 4, 5\}$, $B = \{1, 3, 5\}$, $C = \{3, 5, 7, 11\}$, $D = \{2, 4, 8, 10\}$

- ☒ 1 $\{1, 2, 3, 5, 6, 7, 8, 9, 10\}$
- ☒ 2 $\{1, 2, 3, 4, 5, 6, 7, 8, 10\}$
- ☒ 3 $\{1, 2, 3, 4, 5, 6, 7, 8, 10, 11\}$
- ☒ 4 $\{1, 2, 3, 4, 6, 7, 8, 10, 11\}$

EXERCISE – 1.3

Question

Make correct statements by filling in the symbols $\subset$ or $\not\subset$ in the blank spaces:

- (i) $\{2, 3, 4\} \subset \{1, 2, 3, 4, 5\}$
- (ii) $\{a, b, c\} \not\subset \{b, c, d\}$
- (iii) $\{x : x \text{ is a student of Class XI of your school}\} \subset \{x : x \text{ student of your school}\}$
- (iv) $\{x : x \text{ is a circle in the plane}\} \not\subset \{x : x \text{ is a circle in the same plane with radius 1 unit}\}$
- (v) $\{x : x \text{ is a triangle in a plane}\} \not\subset \{x : x \text{ is a rectangle in the plane}\}$
- (vi) $\{x : x \text{ is an equilateral triangle in a plane}\} \subset \{x : x \text{ is a triangle in the same plane}\}$
- (vii) $\{x : x \text{ is an even natural number}\} \subset \{x : x \text{ is an integer}\}$

Question

Examine whether the following statements are true false

- ✓ (i) $\{a, b\} \not\subset \{b, c, a\}$ (False)
- ✓ (ii) $\{a, e\} \subset \{x : x \text{ is a vowel in the English alphabet}\}$ True.
 $= \{a, e, i, o, u\}$
- (iii) $\{1, 2, 3\} \subset \{1, 3, 5\}$ (False)
- (iv) $\{a\} \subset \{a, b, c\}$ (True)
- (v) $\{a\} \in \{a, b, c\}$ (False)
- (vi) $\{x : x \text{ is an even natural number less than } 6\} \subset \{x : x \text{ is a natural number which divides } 36\}$
 (True)

Question



Let $A = \{1, 2, \{3, 4\}, 5\}$. Which of the following statements are incorrect and why?

~~(i)~~ $\{3, 4\} \subset A$

$\checkmark$ (ii) $\{3, 4\} \in A$

$\checkmark$ (iii) $\{\{3, 4\}\} \subset A$

$\checkmark$ (iv) $1 \in A$

~~(v)~~ $1 \subset A$

$\checkmark$ (vi) $\{1, 2, 5\} \subset A$

~~(vii)~~ $\{1, 2, 5\} \in A$

~~(viii)~~ $\{1, 2, 3\} \subset A$

~~(ix)~~ $\phi \in A$

$\checkmark$ (x) $\phi \subset A$

~~(xi)~~ $\{\phi\} \subset A$

$\{\phi\} \subset A$
 $\Rightarrow \phi \in A.$

(i) Incorrect $\{3, 4\} \in A.$
 $\Rightarrow \{\{3, 4\}\} \subset A$

Question



Write down all the subsets of the following sets

(i) $\{a\} \longrightarrow \phi, \{a\}$

(ii) $\{a, b\} \longrightarrow \phi, \{a\}, \{b\}, \{a, b\}$

(iii) $\{1, 2, 3\} \longrightarrow \phi, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}$

(iv) $\phi \longrightarrow \phi$

Question



Write the following as intervals:

✓(i) $\{x : x \in \mathbb{R}, -4 < x \leq 6\} = (-4, 6]$

✓(ii) $\{x : x \in \mathbb{R}, -12 < x < -10\} = (-12, -10)$

✓(iii) $\{x : x \in \mathbb{R}, 0 \leq x < 7\} = [0, 7)$

✓(iv) $\{x : x \in \mathbb{R}, 3 \leq x \leq 4\} = [3, 4]$

Question



Write the following intervals in **set-builder form**:

$$(i) \quad (-3, 0) = \{x \in \mathbb{R} : -3 < x < 0\}$$

$$(ii) \quad [6, 12] = \{x \in \mathbb{R} : 6 \leq x \leq 12\}$$

$$(iii) \quad (6, 12] = \{x \in \mathbb{R} : 6 < x \leq 12\}$$

$$(iv) \quad [-23, 5) = \{x \in \mathbb{R} : -23 \leq x < 5\}.$$

Question

What universal set(s) would you propose for each of the following:

- (i) The set of right triangles.
- (ii) The set of isosceles triangles.

(Sol) $\cup$ = Set of Triangles.

Question

Given the sets $A = \{1, 3, 5\}$, $B = \{2, 4, 6\}$ and $C = \{0, 2, 4, 6, 8\}$, which of the following may be considered as universal set (s) for all three sets A, B and C

- ~~(i)~~ $\{0, 1, 2, 3, 4, 5, 6\}$ ✗
- ~~(ii)~~ ϕ ✗
- ~~(iii)~~ $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$
- ~~(iv)~~ $\{1, 2, 3, 4, 5, 6, 7, 8\}$

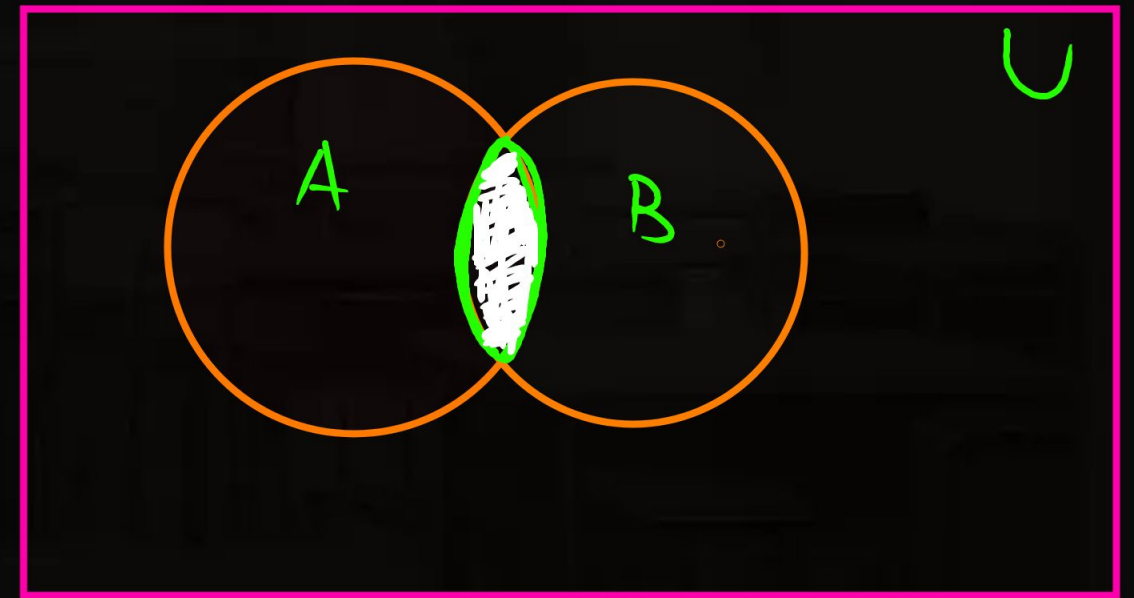
Venn Diagram

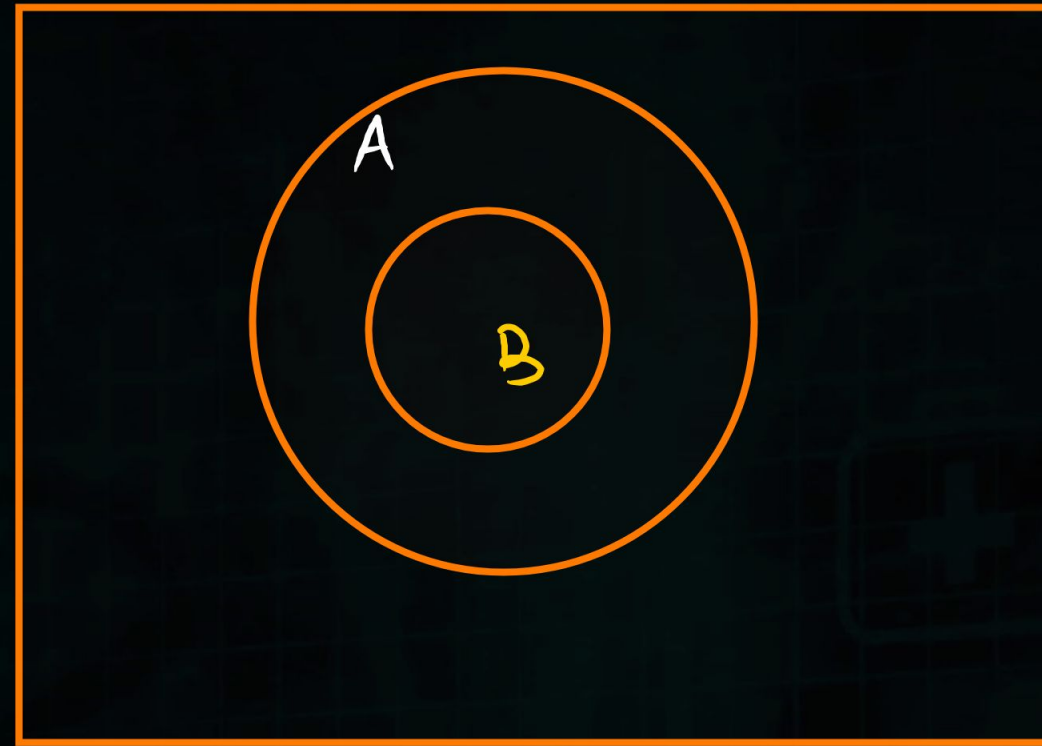
- Pictorial representation of all the possible relationships between sets w.r.t. the universal set
- Universal set is shown by rectangle
- Sets are shown by closed curves (circles)

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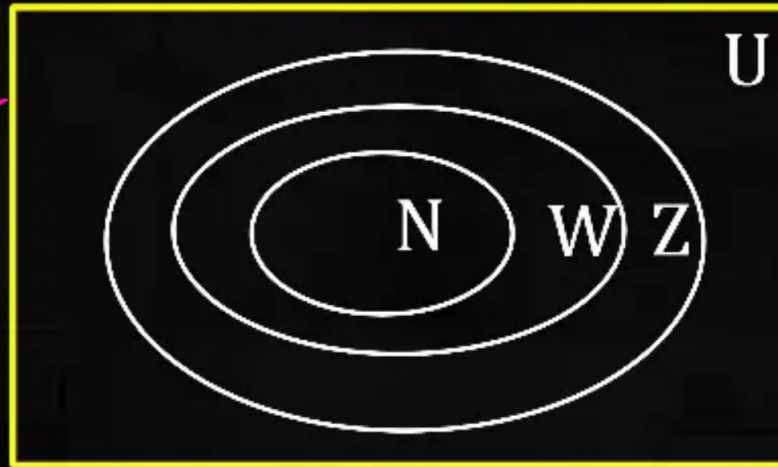
B is a subset of A

QUESTION

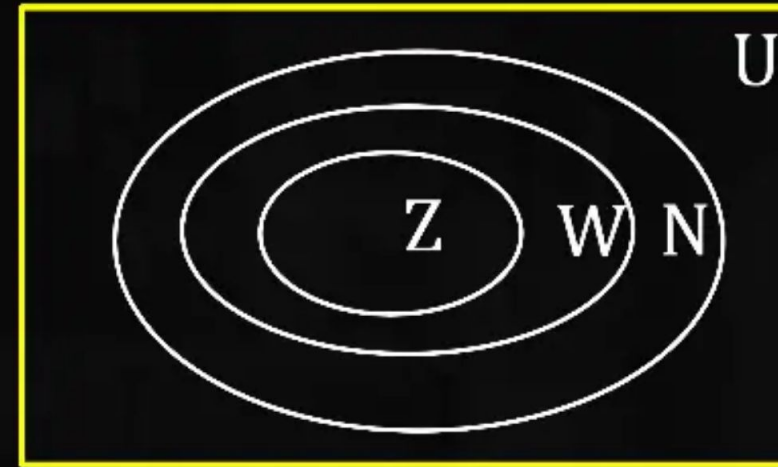
The set of natural numbers is a subset of whole numbers which is a Subset of Integers.

Then, correct representation of it through Venn Diagram is :

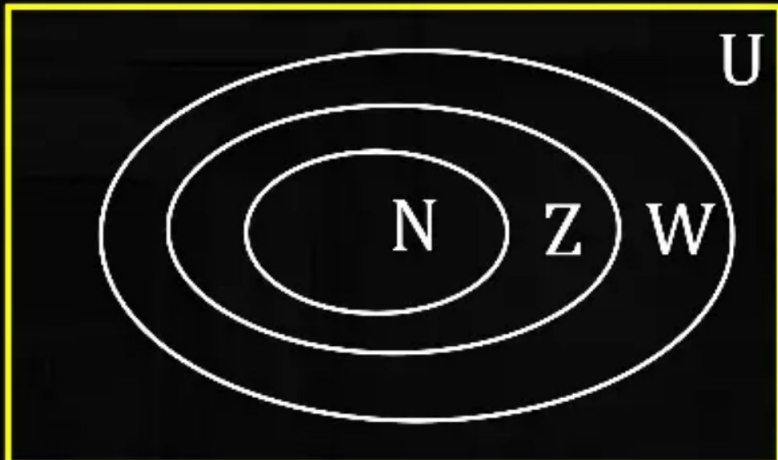
~~1~~



~~2~~



~~3~~



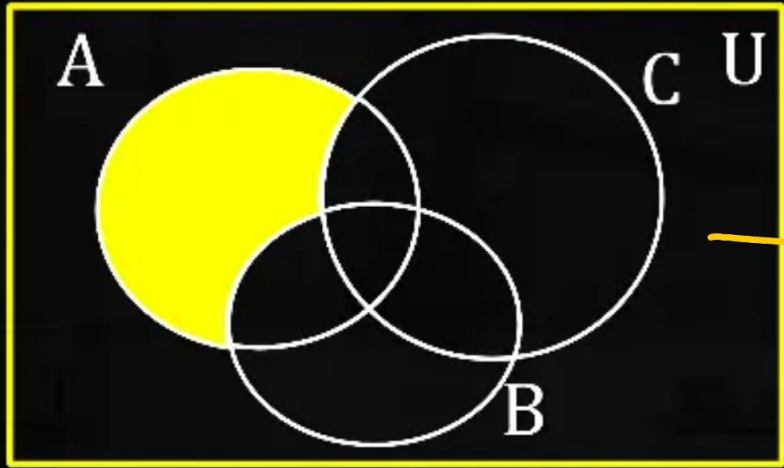
~~4~~

None of These

QUESTION

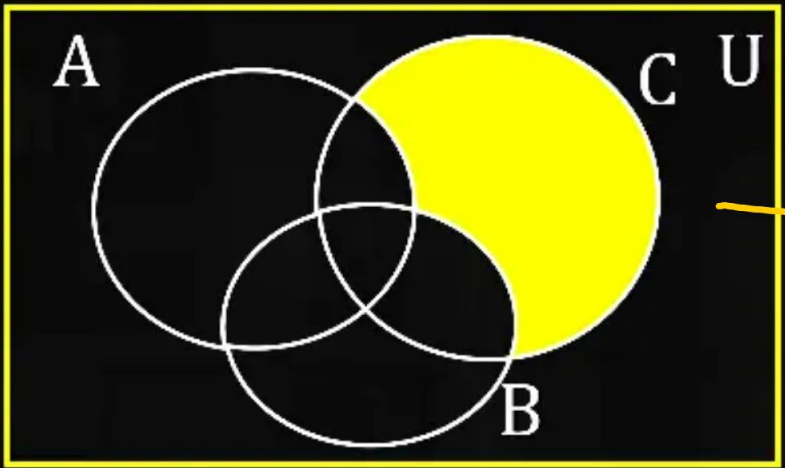
Out of the following Venn Diagrams, which represents **only A**, given three sets A, B and C ?

1



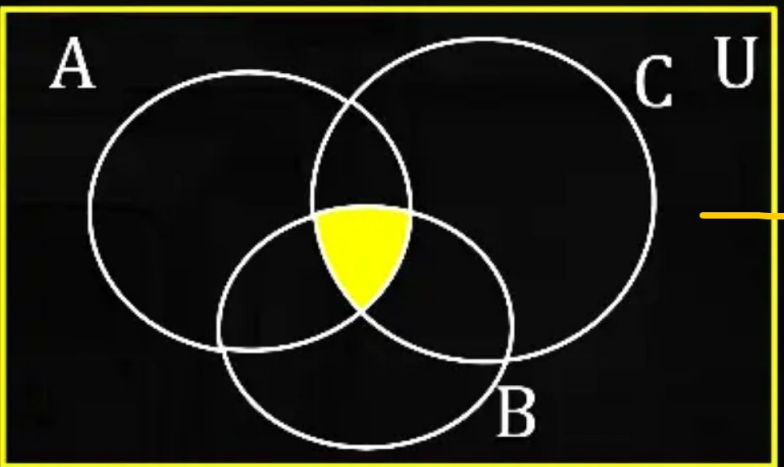
→ only A

~~**2**~~



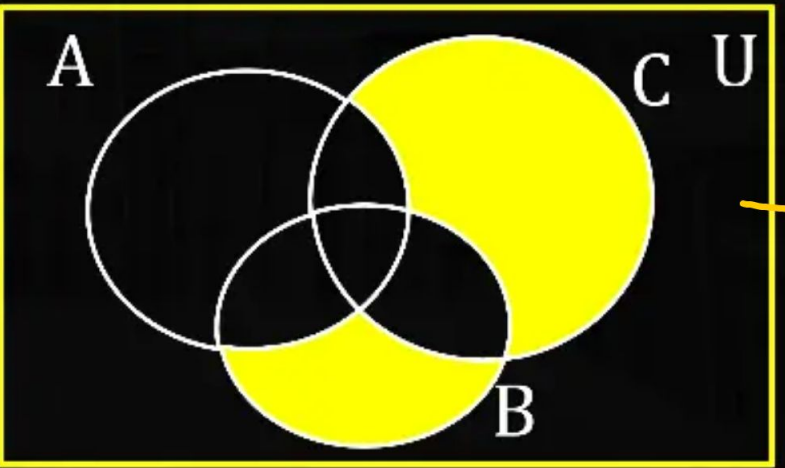
→ only C

~~**3**~~



→ A, B, C

~~**4**~~

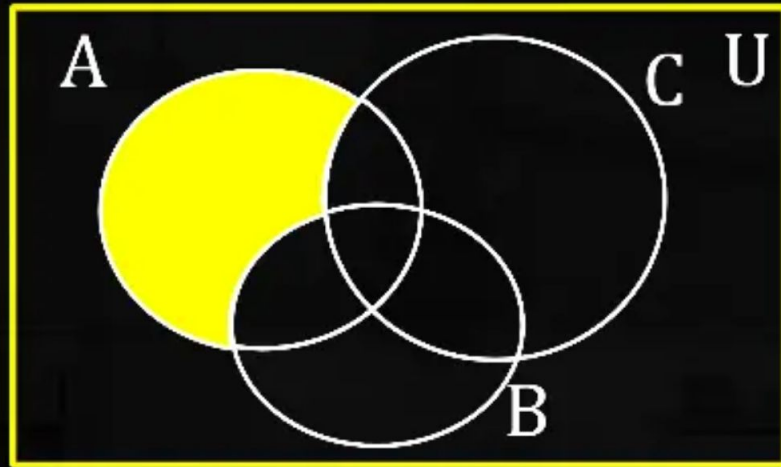


→ only B & only C

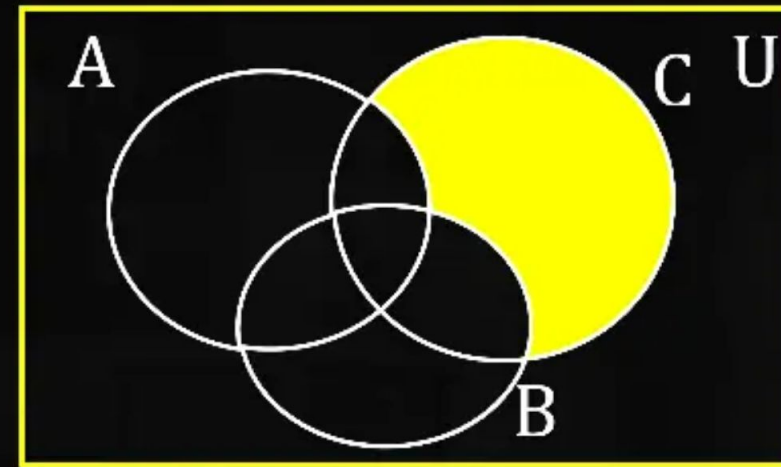
QUESTION

Out of the following Venn Diagrams which represents the region common to all the sets A, B and C?

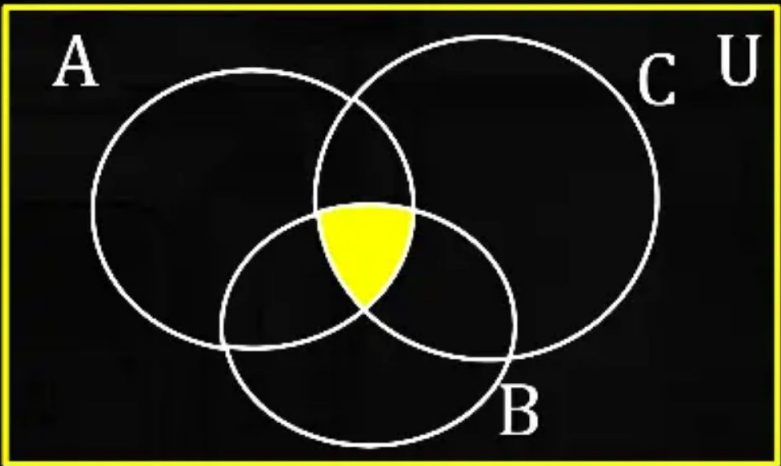
~~1~~



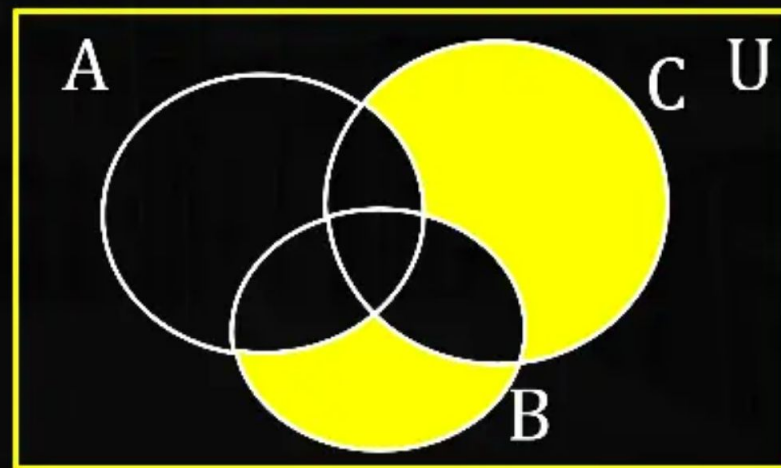
~~2~~



~~3~~



~~4~~



Thank
You